

Food Delivery & Last-Mile Operations

Analysis

1. Project Overview

Developed an interactive food delivery analytics dashboard using Python, Pandas, Matplotlib, Seaborn, and Power BI to analyze order performance, revenue trends, cancellations, restaurant efficiency, rider wait times, and delivery operations. The project helps identify operational bottlenecks, customer behavior patterns, revenue-driving locations, and key factors affecting delivery performance. The insights support data-driven decision-making, improve operational efficiency, reduce cancellations, and enhance customer satisfaction.

Project Objectives

- To analyze food delivery order data effectively.
- To monitor order performance and revenue generation.
- To identify top-performing subzones and restaurants.
- To analyze cancellation trends and their root causes.
- To evaluate kitchen preparation time and rider wait time efficiency.
- To identify operational bottlenecks affecting delivery performance.
- To understand customer ordering patterns and spending behavior.
- To provide interactive and meaningful visual insights using Python and Power BI.

2. Data Sources

- **Source Description and Timeline:**

<https://www.kaggle.com/datasets/sujalsuthar/food-delivery-order-history-data>

- **Domain:** E-commerce

3. Problem Statement

The objective of this project is to analyze food delivery operations and identify key factors influencing revenue, order fulfillment, cancellations, restaurant performance, and rider efficiency. Through descriptive analytics and data visualization, the project provides insights that support operational improvements and better business decision-making.

4. Attribute (Column /Features) Details:

- **Restaurant ID** - Unique identifier assigned to each restaurant
- **Restaurant name** - Name of the restaurant fulfilling the order.
- **Subzone** - Specific delivery area or locality within a city.
- **City** - City where the order was placed and delivered.
- **Order ID** - Unique identifier for each order.
- **Order Placed At** - Timestamp indicating when the customer placed the order.
- **Order Status** - Current status of the order (Delivered, Cancelled, Rejected, etc.).
- **Delivery** - Indicates whether delivery service was used for the order.
- **Distance** - Distance between restaurant and delivery location (in km or meters).
- **Items in order** - Total number of food items included in the order.

- **Instructions** - Special delivery or food preparation instructions provided by the customer.
- **Discount construct** - Type or structure of discount applied to the order.
- **Bill subtotal** - Total food amount before applying discounts and additional charges.
- **Packaging charges** - Additional charges applied for food packaging.
- **Restaurant discount (Promo)** - Promotional discount offered by the restaurant.
- **Restaurant discount (Flat offs, Freebies & others)** - Discounts such as flat offers, freebies, or special restaurant promotions.
- **Gold discount** - Discount provided to premium/gold membership customers.
- **Brand pack discount** - Discount applied through brand-specific promotional packs.
- **Total** - Final payable amount after all discounts and charges.
- **Rating** - Customer rating given for the order/service.
- **Review** - Written customer feedback or review for the order experience.
- **Cancellation / Rejection reason** - Reason why the order was cancelled or rejected.
- **Restaurant compensation (Cancellation)** - Compensation amount paid by the restaurant for cancelled orders.
- **Restaurant penalty (Rejection)** - Penalty charged to the restaurant for rejected orders.
- **KPT duration (minutes)** - Kitchen Preparation Time taken to prepare the order.
- **Rider wait time (minutes)** - Time the delivery rider waited at the restaurant before pickup.
- **Order Ready Marked** - Timestamp or status indicating when the order was marked ready for pickup.
- **Customer complaint tag** - Category/tag describing customer complaints related to the order.

- **Customer ID** - Unique identifier assigned to each customer.

6. Data Pre-Processing (Python / Pandas)

Tasks Performed:

- Removed duplicate records to ensure data consistency.
- Handled missing values in KPT duration, Rating, Rider Wait Time, and Distance columns.
- Converted data types (e.g., Restaurant ID to string, Distance and KPT to numeric).
- Replaced missing KPT values using restaurant-wise average KPT based on business rules.
- Filled missing ratings using customer complaint information and restaurant average ratings.
- Standardized categorical values and cleaned text fields.
- Detected and treated outliers using the Interquartile Range (IQR) method.
- Renamed columns for better readability (e.g., Distance → Distance in km).
- Created clean and analysis-ready datasets for EDA and dashboard development.

SCREENSHOT OF DATA CLEANING CODE

 `print(df.columns)`

```
... Index(['Restaurant ID', 'Restaurant name', 'Subzone', 'City', 'Order ID',
        'Order Placed At', 'Order Status', 'Delivery', 'Distance',
        'Items in order', 'Instructions', 'Discount construct', 'Bill subtotal',
        'Packaging charges', 'Restaurant discount (Promo)',
        'Restaurant discount (Flat offs, Freebies & others)', 'Gold discount',
        'Brand pack discount', 'Total', 'Rating', 'Review',
        'Cancellation / Rejection reason',
        'Restaurant compensation (Cancellation)',
        'Restaurant penalty (Rejection)', 'KPT duration (minutes)',
        'Rider wait time (minutes)', 'Order Ready Marked',
        'Customer complaint tag', 'Customer ID'],
        dtype='object')
```

```
print(df.head())
```

```
... Restaurant ID Restaurant name Subzone City Order ID \
0 20320607 Swaad Sector 4 Delhi NCR 6168884918
1 20320607 Swaad Sector 4 Delhi NCR 6170707559
2 20320607 Swaad Sector 4 Delhi NCR 6169375019
3 20320607 Swaad Sector 4 Delhi NCR 6151677434
4 20320607 Swaad Sector 4 Delhi NCR 6167540897

Order Placed At Order Status Delivery Distance \
0 11:38 PM, September 10 2024 Delivered Zomato Delivery 3km
1 11:34 PM, September 10 2024 Delivered Zomato Delivery 2km
2 03:52 PM, September 10 2024 Delivered Zomato Delivery <1km
3 03:45 PM, September 10 2024 Delivered Zomato Delivery 2km
4 03:04 PM, September 10 2024 Delivered Zomato Delivery 2km

Items in order ... Rating Review \
0 1 x Grilled Chicken Jamaican Tender, 1 x Grill... ... NaN NaN
1 1 x Peri Peri Fries, 1 x Fried Chicken Angara ... ... NaN NaN
2 1 x Bone in Peri Peri Grilled Chicken ... ... NaN NaN
3 1 x Fried Chicken Ghostbuster Tender, 1 x Anga... ... 4.0 NaN
4 1 x Peri Peri Krispers, 1 x Fried Chicken Anga... ... NaN NaN
```

```
print(df.info())
```

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 21321 entries, 0 to 21320
Data columns (total 29 columns):
# Column Non-Null Count Dtype
---
0 Restaurant ID 21321 non-null int64
1 Restaurant name 21321 non-null object
2 Subzone 21321 non-null object
3 City 21321 non-null object
4 Order ID 21321 non-null int64
5 Order Placed At 21321 non-null object
6 Order Status 21321 non-null object
7 Delivery 21321 non-null object
8 Distance 21321 non-null object
9 Items in order 21321 non-null object
10 Instructions 720 non-null object
11 Discount construct 15823 non-null object
12 Bill subtotal 21321 non-null float64
13 Packaging charges 21321 non-null float64
14 Restaurant discount (Promo) 21321 non-null float64
15 Restaurant discount (Flat offs, Freebies & others) 21321 non-null float64
16 Gold discount 21321 non-null float64
17 Brand pack discount 21321 non-null float64
18 Total 21321 non-null float64
19 Rating 21321 non-null float64
```

```
print(df.describe())
```

```
...      Restaurant ID      Order ID  Bill subtotal  Packaging charges \
count      2.132100e+04  2.132100e+04    21321.000000    21321.000000
mean      2.074413e+07  6.354622e+09      750.076838      32.564592
std       2.447193e+05  1.230263e+08     498.759428     22.235898
min       2.032061e+07  6.086767e+09      50.000000      0.000000
25%      2.063570e+07  6.250751e+09     459.000000     18.450000
50%      2.065987e+07  6.357715e+09     629.000000     28.450000
75%      2.088265e+07  6.456827e+09     899.000000     39.950000
max       2.152306e+07  6.573392e+09    16080.000000    603.000000

      Restaurant discount (Promo) \
count                21321.000000
mean                  65.091816
std                   85.401604
min                   0.000000
25%                  0.000000
50%                  80.000000
75%                 100.000000
max                 4020.000000

      Restaurant discount (Flat offs, Freebies & others)  Gold discount \
count                21321 0000000    21321 0000000
```

```
print(df.duplicated().sum())
```

```
... 0
```

```
# Replacing null value by taking account of customer complaint
for i in df.index:

    if pd.isnull(df.loc[i, 'Rating']):

        # If complaint tag is null
        if pd.isnull(df.loc[i, 'Customer complaint tag']):

            df.loc[i, 'Rating'] = 5

        # Otherwise fill with restaurant average rating
        else:

            avg_rating = df[
                df['Restaurant ID'] == df.loc[i, 'Restaurant ID']
            ]['Rating'].mean()

            df.loc[i, 'Rating'] = round(avg_rating, 1)
```

```
# Replacing null value
df['Instructions']=df['Instructions'].fillna('No Instructions')
```

```
# Replacing null value
df['Discount construct'] = df['Discount construct'].fillna('No Discount')
```

```
# Replacing null value
df['Review'] = df['Review'].fillna('No Command')
```

```
# Replacing null value
df['Cancellation / Rejection reason'] = df['Cancellation / Rejection reason'].fillna('No cancellation')
```

```
# Replacing null value
df['Restaurant compensation (Cancellation)'] = df['Restaurant compensation (Cancellation)'].fillna(0.0)
```

```
# Replacing null value
df['Restaurant penalty (Rejection)'] = df['Restaurant penalty (Rejection)'].fillna(0.0)
```

```
# Replacing null value
df['Customer complaint tag'] = df['Customer complaint tag'].fillna('No complain')
```

```
# Replacing null value
df['Rider wait time (minutes)'] = df['Rider wait time (minutes)'].fillna(0.0)
```

```
▶ # Changing the data type and replacing null value with logic
df['KPT duration (minutes)'] = pd.to_numeric(
    df['KPT duration (minutes)'],
    errors='coerce'
)

for i in df.index:
    if pd.isnull(df.loc[i, 'KPT duration (minutes)']):
        if df.loc[i, 'Cancellation / Rejection reason'] == 'Merchant device issue':

            avg_kpt = df[
                df['Restaurant ID'] == df.loc[i, 'Restaurant ID']
           ]['KPT duration (minutes)'].mean()

            df.loc[i, 'KPT duration (minutes)'] = round(avg_kpt, 2)
        else:
            df.loc[i, 'KPT duration (minutes)'] = 0.0
```

```

▶ #Using IQR method for outlier in wait time
Q1 = df['Rider wait time (minutes)'].quantile(0.25)
Q3 = df['Rider wait time (minutes)'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

# Create a new dataframe without outliers
df_no_outliers = df[
    (df['Rider wait time (minutes)'] >= lower_limit) &
    (df['Rider wait time (minutes)'] <= upper_limit)
]

print("Original rows:", len(df))
print("Rows after removing outliers:", len(df_no_outliers))
print("Outliers removed:", len(df) - len(df_no_outliers))

```

```

... Original rows: 21321
Rows after removing outliers: 20797
Outliers removed: 524

```

```

# Removing the column
df.drop(columns=['Instructions'], inplace=True)

```

```

#removing and changing data type
df['Distance'] = df['Distance'].str.replace('km', '')

df['Distance'] = pd.to_numeric(df['Distance'], errors='coerce')

```

```

# Renameing the column
df.rename(columns={'Distance': 'Distance in km'}, inplace=True)

```

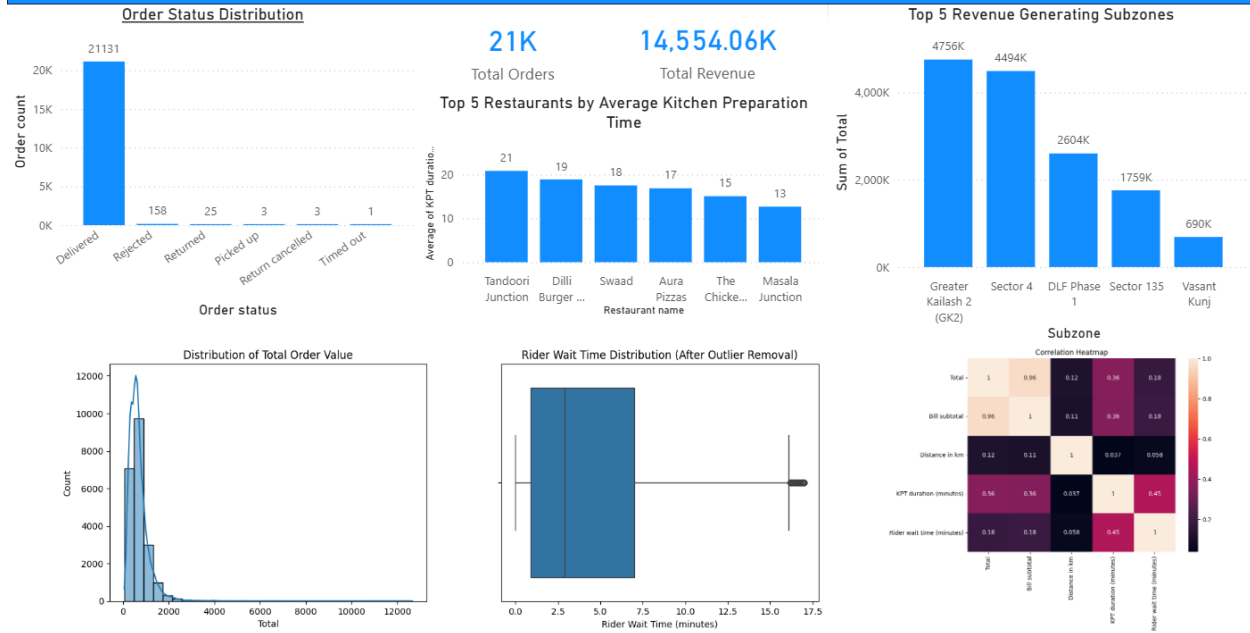
```

▶ # Replacing null value
df['Distance in km'] = df['Distance in km'].fillna(0)

```

SCREENSHOT OF DASHBOARD /REPO

Executive Summary



Dashboard insight:

Excellent Order Fulfillment Performance

Insight:

- 99% orders were successfully delivered. Rejected, returned, cancelled, and timed-out orders account for less than 1% of total orders.

Business Impact:

Indicates strong operational efficiency and high service reliability. Customer order completion rate is exceptionally high.

Recommendation:

- Maintain current fulfillment processes.
- Conduct root-cause analysis on rejected and cancelled orders to further improve service quality.

Revenue is Concentrated in a Few Key Subzones

Insight:

- Greater Kailash 2 (GK2) and Sector 4 generate the highest revenue. These locations contribute a significant share of total revenue despite representing only a few subzones.

Business Impact:

- Revenue dependency exists on specific geographic areas. Any operational issue in these zones could significantly impact revenue.

Recommendation:

- Prioritize customer retention and service quality in GK2 and Sector 4. Expand marketing efforts in other promising subzones to diversify revenue sources.

Kitchen Preparation Time Directly Affects Rider Waiting Time

Insight:

- Correlation between Kitchen Preparation Time (KPT) and Rider Wait Time = 0.45 (moderate positive relationship). Restaurants with longer preparation times create longer rider waits.

Business Impact:

- Increased rider idle time can reduce delivery efficiency.
- Longer waits may increase cancellation risk and operational costs.

Recommendation:

- Monitor restaurants with high KPT (e.g., Tandoori Junction).
- Introduce kitchen performance KPIs and preparation-time benchmarks. Optimize staffing during peak hours.

Customer Spending is Highly Skewed

Insight:

- Most orders fall in the low-to-medium value range. A small number of high-value orders create a long revenue tail.

Business Impact:

- Revenue is primarily driven by frequent regular customers.
- Premium customers represent an important growth opportunity.

Recommendation:

- Create loyalty programs for high-spending customers.
- Use upselling and bundle offers to increase Average Order Value (AOV).

Distance Has Minimal Impact on Operational Performance

Insight:

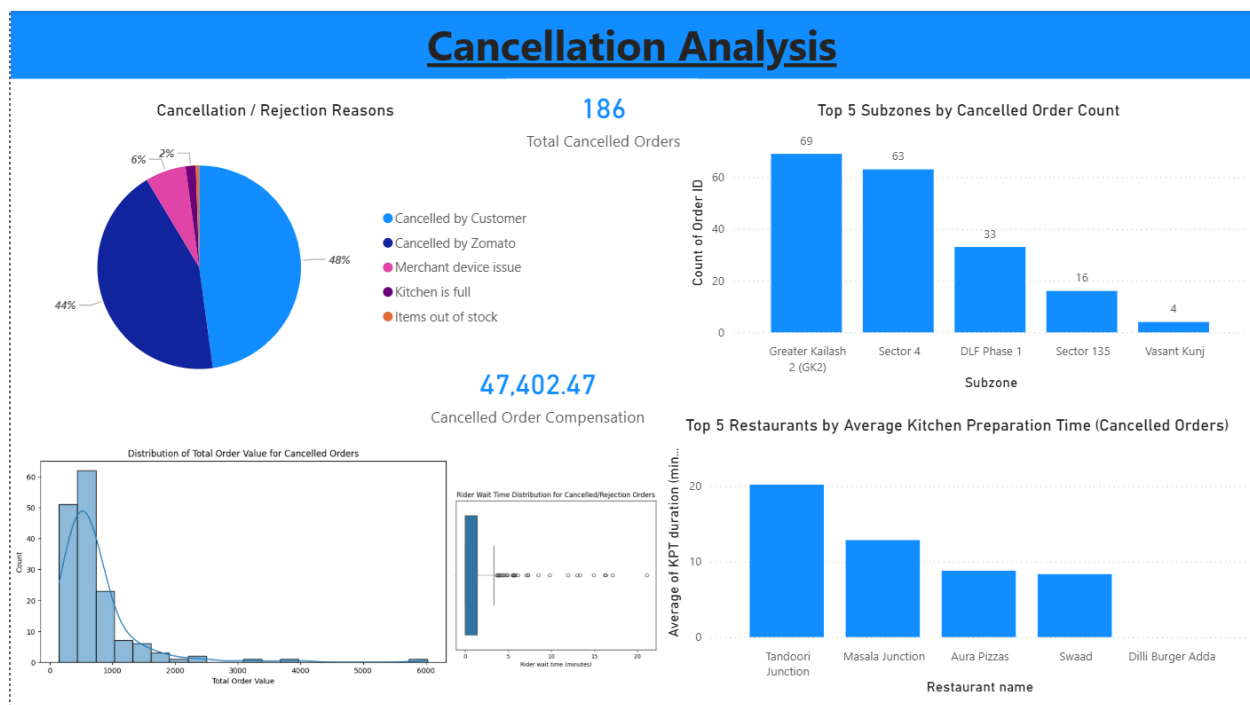
- Distance shows very weak correlation with: Kitchen Preparation Time (0.04), Rider Wait Time (0.06), Order Value (0.12).

Business Impact:

- Operational delays are not primarily caused by delivery distance.
- Internal restaurant processes have a greater impact on performance.

Recommendation:

- Focus improvement efforts on kitchen operations and rider allocation rather than delivery distance optimization.



Dashboard insight:

Customer and Zomato Cancellations Drive Most Order Losses

Insight

- Cancelled by Customer (48%) and Cancelled by Zomato (44%) together account for 92% of all cancellations.
- Operational issues such as merchant device problems (6%), kitchen capacity (2%), and stock issues (<1%) contribute very little.

Business Impact

- Most cancellations are not caused by inventory or restaurant capacity issues.

- Customer behavior and platform-level operations are the primary drivers.

Recommendation

- Improve delivery time estimates and order tracking.
- Investigate reasons behind Zomato-initiated cancellations.
- Implement proactive customer communication to reduce voluntary cancellations.

Cancellations Are Concentrated in High-Revenue Subzones

Insight

- Greater Kailash 2 (69 cancellations) and Sector 4 (63 cancellations) have the highest cancellation counts.
- These same areas are also among the top revenue-generating subzones.

Business Impact

- Revenue-generating locations are also experiencing the highest order losses.
- Even a small reduction in cancellations in these zones could significantly increase revenue.

Recommendation

- Conduct zone-level root cause analysis in GK2 and Sector 4.
- Improve delivery partner availability and restaurant coordination in these areas.
- Monitor cancellation KPIs by subzone.

Kitchen Delays May Be Contributing to Cancellations

Insight

- Tandoori Junction (20.17 min) has the highest average Kitchen Preparation Time (KPT) among cancelled orders.
- Other restaurants such as Masala Junction also show relatively high preparation times.

Business Impact

- Longer preparation times can increase customer waiting time and cancellation probability.
- Delays may also increase rider idle time and operational costs.

Recommendation

- Set preparation time benchmarks for restaurants.
- Improve kitchen workflow and staffing during peak hours.

- Alert restaurants when preparation times exceed thresholds.

Most Cancelled Orders Are Low-to-Medium Value Orders

Insight

- The cancelled order value distribution is heavily concentrated below ₹1,000.
- Only a few cancelled orders exceed ₹3,000–₹6,000.

Business Impact *Most cancellations come from regular customers rather than premium orders.

- However, high-value cancelled orders represent disproportionate revenue loss.

Recommendation

- Focus on reducing cancellation frequency among regular orders.
- Provide priority support for high-value orders to prevent revenue leakage.

Rider Wait Times Are Generally Not the Main Cause of Cancellations

Insight

- Most rider wait times for cancelled/rejected orders are relatively low.
- Only a few outlier cases show significantly longer waits.

Business Impact

- Rider delays do not appear to be the primary driver of cancellations.
- Other factors such as customer decisions, platform issues, and kitchen delays have a greater impact. Recommendation
- Continue monitoring rider performance. Prioritize improvements in kitchen preparation and order management rather than rider allocation.

Final Overall Insights:

- The descriptive analysis reveals a highly efficient food delivery operation with over 21,000 orders processed, more than 99% successful deliveries, and ₹14.55 million in revenue.
- Revenue is concentrated in key subzones such as GK2 and Sector 4.

- while customer and platform-driven cancellations remain the primary source of order loss.
- Restaurant preparation time emerges as the most influential operational factor, directly impacting rider wait times and potentially contributing to cancellations. Overall, the business demonstrates strong operational performance, with future improvements focused on reducing cancellations, optimizing kitchen efficiency, and strengthening service quality in high-demand locations.

Conclusion:

The analysis revealed important trends in order performance, revenue, cancellations, and delivery operations. The insights highlight key business drivers and operational improvement areas, providing valuable support for informed decision-making and future business growth.